int a = 10;
↳ Memory [2000] → Address
↓
10

Address Operator = &a;
Pointer → int *ptr = &a;   (ptr: 3000)
↳ &ptr = 3000

Value of a → a, *ptr

int **dPtr = &ptr;

ptr → address of a
*ptr → value of a

***tPtr = &dptr;
[***tptr] = 10

***p → p → p → (a)

dptr → address of ptr
*dptr → ptr → value
**dptr → [2000] → (10)

C/C++|Java: Array → Homogeneous Collection of Data
Same data
List[Items]

Python data = [1, 2.5, True, 'Java'] Heterogeneous
i f b S

int arr[] = {2, 9, 6, 1, 0, 5}; → Element
0 1 2 3 4 5 → Indexes / Indices
4th
arr[elements]   3rd → arr[3]   Last arr[size - 1]
Size = sizeof(arr) / sizeof(arr[0])   Last arr[n - 1]

int arr[] = {2, 4, 6, 8, 0};   4 bytes
2000, 2004, 2008, 2012, 2016
↓
var

* The name of the array points to the address of the 1st element of the array.
arr → 2000
Array Pointer → arr → 1st element pointer.

int arr[] = [2, 6, 1, 9, 5, 4]
*arr, arr+1, arr+2
2000, 2004, 2008, 2012, 2016, 2020
1, 9, 5, 4
→ [arr + 0] i = 0
2000
arr + 1
int
0
arr + i
2000 4bytes
2004 4bytes
(arr+1)
2000 + 4
arr + (n-1)

(0 - 6)
*arr + i
2 + 0 = 2
2 + 1 = 3
2 + 2 = 4

*(2000) = 2 > 6,
*(arr + i)
arr + 0
(arr + 1) address
arr = 2000 [for int i = 0]
0
↓
4bytes
4
2000+4

* Pattern: → 2D
i, j
for
for

Searching & Sorting Algos
* Linear
* Binary
* Recursive BS
* Interpolation
* Jump Search

* Bubble
Selection
Insertion
Heap

Tree
NLDS
Heap

* Identity Matrix:
H.W.
r == c → 1
0
[4PM] *

1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

* Sparse Matrix:
No. of elements:
ctr → 10
4x5
r x c = 20
if {ctr > ((r*c)/2)}
SM
NSM

1	2	0	1	0
0	3	6	1	2
0	0	0	0	0
0	1	0	0	1

11 → 20
SM

~~Sparse Matrix~~ ✓
No of zeros → (10)
No. of elements → [20]
(10)
More than half ?

* Transpose of a Matrix:
Trick
* row becomes col
* col becomes row

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

r → 1 2 3 4
r → 5 6 7 8

1 5
2 6
3 7
4 8
↑

UTM:
1 2 3 4
6 7 8
11 12
16

* Identity
HW * Transpose
* Upper Triangle
* Lower Traingle

LTM
1
5 6
9 10 11
13 14 15 16

* Dynamic Memory Allocation:
C → malloc( )
→ calloc( )
C++ → realloc( )
→ free( )
C + C++ = Java;

C++
new delete → C o = n C();
DMA in C++
* Why is DMA needed?

11:00 to 1:00
2:00